

An Accurate and Efficient Stochastic Solver for Transient Thermal Analysis With Mixed Boundary Conditions

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Abstract—Transient thermal analysis captures the high accuracy and time-varying temperature distribution of chip components. However, during the early thermal design iterations of 3-D ICs and chiplets, time-consuming transient thermal simulations led to prolonged waiting times for engineers. Existing widely adopted deterministic methods or innovative neural network-based (NN-based) methods face significant challenges in reducing the thermal simulation time when geometric structures change iteratively. With prior knowledge of the chip’s regions of interest, localized stochastic methods may present a promising solution. In this work, we propose a novel high-precision and efficient stochastic solver for transient thermal analysis of ICs, based on the probabilistic solution of the heat equation obtained from the Feynman–Kac formula. As a field solver for the heat equation, the proposed method adopts the *spacetime walk-on-spheres* (WOS) scheme and enhances it with the novel boundary treatment integrated to guarantee the accuracy of addressing mixed boundary conditions, particularly Robin boundary conditions. A time-dependent fixed random walk is also combined to handle heat-source regions. Experimental results show that the proposed solver maintains high accuracy with average absolute error within 0.27 K and maximum absolute error within 1 K, achieves a speedup of 26.8–211 $\times$ at a single point compared to the 3D-ICE simulator of similar accuracy, has general geometry applicability as a field solver, and exhibits excellent parallel efficiency.

Index Terms—Feynman–Kac formula, random walk method, Robin boundary conditions, spacetime walk-on-spheres (WOS) scheme, transient thermal analysis.

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I. INTRODUCTION

AS THE number of transistors increases rapidly, the rising power density and temperature make the thermal issue a major challenge in IC design, especially for stacked 3-D ICs or chiplets. Elevated temperatures degrade chip performance, generate greater subthreshold leakage power, and affect the long-term reliability and aging of chips [1], [2]. Therefore, accurate and fast thermal analysis of ICs is of vital importance during the early stages of IC design.

Steady-state thermal analysis provides final temperatures for given power maps but cannot capture thermal dynamics. Due to frequent load changes during the operation of modern chips, tracking the temporal evolution of temperatures is critical. Therefore, transient thermal analysis is essential as it enables engineers to identify and address transient thermal issues, thereby ensuring the reliability and performance of devices under various operating conditions.

Current thermal analysis methods roughly fall into three categories: deterministic, neural network-based (NN-based), and stochastic methods. Deterministic methods are numerical field solvers for solving the partial differential equations (PDEs) of heat conduction. Based on global discretization and solving the linear equations, they are extensively adopted in both steady-state and transient thermal analysis, of which finite-element methods (FEMs) and finite-difference methods (FDMs) are typical representatives. FEMs divide the global domain into multiple elements and select an approximate function for each element. The governing equations for each element are then formulated and assembled into a linear system of equations to be solved. FEMs can achieve very high accuracy through extremely fine discretization, and are generalizable to complex structures with unstructured mesh [3]. Therefore, it is widely integrated into commercial simulators like ANSYS [4] and COMSOL [5]. FDMs are simpler to implement and popular in academia [6], [7], especially 3D-ICE [8], [9], [10] and HotSpot [11]. FDMs utilize the duality between thermal and electric properties to create compact thermal models (CTMs). In CTMs, the chip is represented as an equivalent thermal resistor-capacitor network of thermal nodes and solved in a manner similar to circuit simulation. However, they seem not as flexible as FEMs for complex geometries and boundary conditions [3], [7]. The deterministic methods can achieve high

accuracy; however, they generally incur significant time and memory overhead when performing transient thermal analysis on large-scale chips with high-resolution grids.

NN-based thermal analysis methods have emerged in recent years. With the embedding of physical information, NNs have shown good performance in dealing with steady-state thermal analysis, such as the methods based on the autoencoder [12], the physics-informed NN [13], and the operator learning [14]. However, in transient thermal analysis, label-dependent data-driven methods still remain predominant. Methods based on the generative adversarial learning [15] and the transformer [16] take real-time performance metrics as inputs and output thermal map estimations. The transient solvers in [17] and [18] utilize the NN to learn the thermal response of a unit heat source or power tile at a given time step, and need postoperations to get temperatures. The method in [19] uses a deep autoencoder with local and global stabilizing techniques to mitigate error accumulation, and the method in [20] proposes an autoregressive operator-learning framework with a Fourier neural operator for multifidelity 3-D-IC thermal analysis. Generally, these NN-based methods are not field solvers because they don't directly solve the heat equation. Consequently, they are highly structure-dependent—changes in chip geometry demand time-consuming, resource-intensive retraining rather than fast inference. Besides, as closed boxes, they offer limited physical interpretability and demand large amounts of high-quality training data with costly acquisition.

To meet the demand for fast transient thermal analysis in large-scale chips during design iterations, a feasible approach is to focus exclusively on regions affected by local design modifications, since engineers are primarily concerned with temperature changes in those areas. Nevertheless, global methods are not so efficient because they cannot directly obtain temperatures of local points without solving the global domain. In contrast, stochastic methods directly target localized solutions, avoid complex global discretization, and are thus well-suited for efficient computation of local temperatures.

Stochastic methods are Monte Carlo-based field solvers that estimate temperatures at points of interest through numerous random walks. In stochastic methods, the precise handling of three types of boundary conditions, i.e., Dirichlet, Neumann, and Robin boundary conditions, for the heat equation has a great impact on the final accuracy. Dirichlet boundary with a given fixed surface temperature is first easily dealt with by path absorption [21]. The Neumann boundary is often empirically considered as an adiabatic boundary in engineering. Robin boundary is also known as a convective boundary with heat transfer toward the outside. Recently, for steady-state thermal analysis, the method in [22] adopts a hybrid random walk algorithm, but fails to handle Robin boundary conditions with a maximum error of more than 20 K [22]. The prevalent Neumann and Robin boundary conditions are finally accurately handled by the path integral random walk solver in [23] with the accurate local time calculation scheme [24] and the Feynman–Kac functional $\hat{e}_c(t)$ [25]. With the extraction of the varying power, the solver is then divided

into an offline precomputation part and an online ultrafast temperature predictor for proactive thermal management [26].

In steady-state thermal analysis, random walks evolve in 3-D space, whereas in transient problems they must evolve in 4-D spacetime (3-D space plus 1-D time). To address this challenge, the 4-D spacetime random-walk framework of Haji-Sheikh and Sparrow [21] can be employed. Nevertheless, its difference-based scheme of probabilistic absorption or reflection at the Robin boundary produces an error of more than 3 K, and it currently cannot handle heat-source regions within the chip. In recent years, the method in [27] focuses on transient coupled conductive and radiative transfers for structures under vacuum without internal heat generation and convection. As far as we are concerned, there are still no stochastic methods for transient thermal analysis that can accurately handle structures containing heat sources and mixed boundary conditions, especially Robin boundary conditions.

In this work, we propose an accurate and efficient stochastic method for transient thermal analysis of 3-D ICs and chiplets. Toward high precision and robustness, the proposed field solver is backed by rigorous mathematical analysis. To mitigate the significant errors in [21], inspired by Ding et al. [24], Simonov [28], and Morillon [29], we introduce a new scheme of boundary local time that markedly reduces these errors. To the best of our knowledge, this is the first work of a high-accuracy stochastic solver for transient thermal problems.

Our contributions are summarized as follows.

- 1) Based on the probabilistic solution of the diffusion equation from the Feynman–Kac formula, a high-precision and efficient stochastic solver is innovatively proposed for transient thermal analysis, which can accurately handle mixed boundary conditions, especially Robin boundary conditions in the heat equation.
- 2) We adopt the *spacetime* walk-on-spheres (WOS) scheme, leveraging the probabilistic constraint relationship between the spatial step and the time step to determine the next spatial position and time interval. The time-dependent fixed random walk is also combined to handle single-layer or multilayer volumetric power maps.
- 3) We propose a new numerical scheme of boundary local time calculation and Feynman–Kac functional updates to improve the accuracy for Neumann and Robin boundary conditions.
- 4) Experimental results show that the proposed solver maintains high accuracy with average absolute error within 0.27 K and maximum absolute error within 1 K, achieves 26.8–211 $\times$ speedup over the state-of-the-art simulator 3D-ICE with similar accuracy at a single point, has general geometry applicability as a field solver, and displays over 90% parallel efficiency over 80 cores.

II. PRELIMINARIES

A. Problem Formulation

Fig. 1 illustrates the formulation of the transient thermal analysis problem of ICs, from the practical chip package [1]

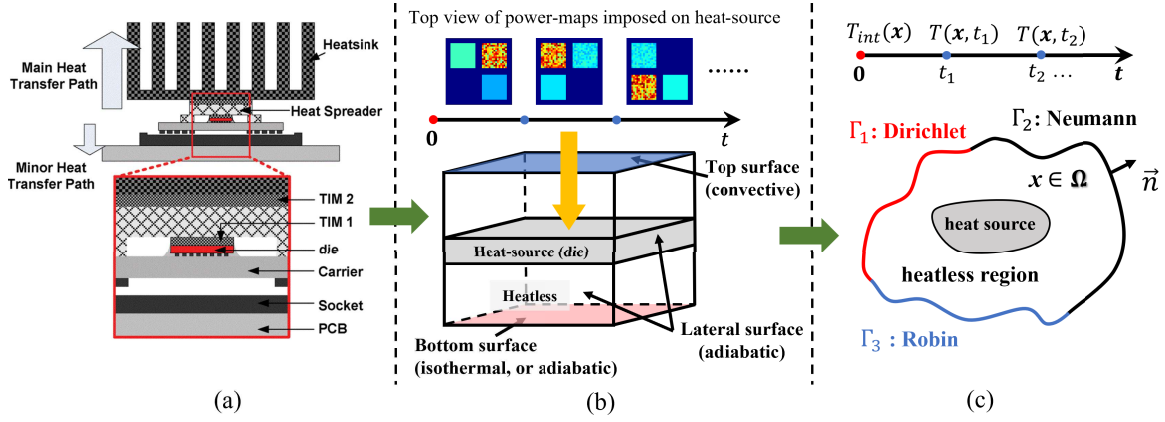


Fig. 1. Illustration of transient thermal analysis for ICs. (a) Practical chip package [1]. (b) Simplified thermal model. (c) Mathematical model.

[Fig. 1(a)], to the simplified thermal model [Fig. 1(b)] and finally to the mathematical model [Fig. 1(c)].

From Fig. 1(a), the die (red color) within the chip package is the heat source where the time-varying temperatures of the interested points are required. Other layers, such as the heat sink, heat spreader, thermal interface material (TIM), carrier, socket, and PCB, are modeled as heatless regions.

From Fig. 1(b), the time-varying power maps are imposed on the heat-source region. Generally, the main heat transfer path is the heat sink (top surface) modeled as a convective (Robin) boundary. The minor heat transfer path is the PCB (bottom surface) conditionally modeled as adiabatic (Neumann), convective (Robin), or isothermal (Dirichlet) boundaries. The four side surfaces, with negligible heat conduction, are usually modeled as adiabatic boundaries.

A 3-D transient heat transfer problem with internal heat generation, as shown in Fig. 1(c), can be formulated as a diffusion equation

$$\frac{\partial T}{\partial t} = \alpha(\mathbf{x}) \nabla^2 T + f(\mathbf{x}, t), \quad \mathbf{x} \in \Omega \subset \mathbb{R}^3 \text{ and } t \geq 0 \quad (1)$$

with initial condition

$$T = T_{\text{ini}}(\mathbf{x}), \quad t = 0 \quad (2)$$

and mixed boundary conditions

$$T = \phi_1(\mathbf{x}, t), \quad \mathbf{x} \in \Gamma_1 \quad (3a)$$

$$\frac{\partial T}{\partial \vec{n}} = \phi_2(\mathbf{x}, t), \quad \mathbf{x} \in \Gamma_2 \quad (3b)$$

$$\frac{\partial T}{\partial \vec{n}} - c(\mathbf{x}, t) T = \phi_3(\mathbf{x}, t), \quad \mathbf{x} \in \Gamma_3 \quad (3c)$$

where $T = T(\mathbf{x}, t)$ is the temperature at $\mathbf{x}$ with time t , $\nabla^2(\cdot)$ is the Laplace operator, and $\vec{n}$ denotes the outer unit normal vector on boundary $\partial\Omega$. Γ_1 , Γ_2 , and Γ_3 are Dirichlet, Neumann, and Robin boundary with the given boundary conditions $\phi_1(\mathbf{x}, t)$, $\phi_2(\mathbf{x}, t)$, and $\phi_3(\mathbf{x}, t)$, respectively. Thermal diffusivity $\alpha(\mathbf{x}) = k/c_p\rho(\mathbf{x})$, where $k(\mathbf{x})$ is thermal conductivity, $c_p(\mathbf{x})$ is specific heat capacity, and $\rho(\mathbf{x})$ is material density. The heat generation term $f(\mathbf{x}, t) = \alpha(\mathbf{x})q(\mathbf{x}, t)/k(\mathbf{x})$, where $q(\mathbf{x}, t)$ is heat generation density. $T_{\text{ini}}(\mathbf{x})$ is the initial temperature distribution in domain Ω .

Note that in this article, the thermal conductivity $k(\mathbf{x})$ can be isotropic or anisotropic, but we are currently unable to handle temperature-dependent $k(T)$. Therefore, we assume $k(\mathbf{x})$ is temperature-independent.

In engineering, Neumann boundary conditions are usually adiabatic boundary conditions with $\phi_2(\mathbf{x}, t) = 0$. Robin boundary conditions (3c) are the main way of heat dissipation in IC thermal problems and can be derived from the convection condition

$$k(\mathbf{x}) \frac{\partial T}{\partial \vec{n}} + h(\mathbf{x}, t)(T - T_{\text{amb}}) = 0 \quad (4)$$

where T_{amb} is the ambient temperature and $h(\mathbf{x}, t)$ is the convective heat transfer coefficient at Robin boundary. Therefore, we have $c(\mathbf{x}, t) = -h(\mathbf{x}, t)/k(\mathbf{x})$ and $\phi_3(\mathbf{x}, t) = h(\mathbf{x}, t)/k(\mathbf{x})T_{\text{amb}}$ in (3c).

For multiple materials with different thermal conductivities k , the heat transfer at the interface of different materials is

$$\begin{aligned} T_{W-} &= T_{W+} \\ -k_{W-} \frac{\partial T_{W-}}{\partial \vec{n}} &= -k_{W+} \frac{\partial T_{W+}}{\partial \vec{n}} \end{aligned} \quad (5)$$

where $W-$ and $W+$ are the two sides of the interface, k_{W-} and k_{W+} are corresponding thermal conductivity, and T_{W-} and T_{W+} are corresponding temperature. The interface condition demonstrates that temperature and heat flux are continuous through the interface. This article considers only perfect contact.

For the steady-state analysis, $\partial T/\partial t = 0$, and the PDE (1) degenerates into the steady-state heat equation

$$-\alpha(\mathbf{x}) \nabla^2 T(\mathbf{x}) = f(\mathbf{x}), \quad \mathbf{x} \in \Omega \quad (6)$$

where all parameters and boundary conditions are time-independent.

B. Stochastic Thermal Analysis Methods

The idea of existing steady-state stochastic thermal analysis methods [23], [26] is illustrated in Fig. 2. They leverage the Feynman–Kac formula to transform the heat equation into the single-point temperature probability solution in the form of path integrals. Then, through extensive Monte Carlo

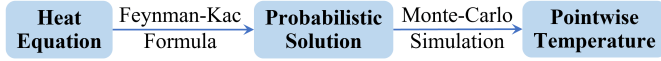


Fig. 2. Idea of stochastic thermal analysis methods.

simulations (random walks), an estimated value of the single-point temperature probability solution can be obtained. In actual chip thermal analysis, with the coordinates of the points of interest as input, the stochastic methods can obtain their temperature solutions point by point through extensive random walks.

C. Probabilistic Solution for Steady-State Heat Equation

The Feynman–Kac formula establishes a link between PDEs and stochastic processes, and gives the probabilistic solution [23], [29] for the steady-state heat equation (6) as

$$T(\mathbf{x}) = E^{\mathbf{x}} \left\{ \hat{e}_c(t_{\Gamma_1}) \phi_1(X_{t_{\Gamma_1}}) \right\} + E^{\mathbf{x}} \left\{ \int_0^\infty \hat{e}_c(t) \phi_{2,3}(X_t) dL(t) \right\} + E^{\mathbf{x}} \left\{ \int_0^\infty \hat{e}_c(t) f(X_t) dt \right\} \quad (7)$$

$$\hat{e}_c(t) = e^{\int_0^t c(X_s) dL(s)} \quad (8)$$

where X_t is the standard reflecting Brownian motion (SRBM) [30] starting at $\mathbf{x}$, $E^{\mathbf{x}}$ is the mathematical expectation, and $\hat{e}_c(t)$ is the Feynman–Kac functional [25] defined on the Robin boundary with the position-dependent coefficient $c(X_t)$. t_{Γ_1} is the time when the path first hits the Dirichlet boundary. $\phi_{2,3}(\mathbf{x})$ denotes Neumann and Robin boundary conditions, $\phi_2(\mathbf{x})$ and $\phi_3(\mathbf{x})$.

The local time $L(t)$ can be expressed as [24], [30], [31]

$$L(t) \equiv \lim_{\epsilon \rightarrow 0} \frac{\int_0^t I_{D_\epsilon}(X_s) ds}{2\epsilon} \quad (9)$$

where D_ϵ is a strip region in the domain containing $\partial\Omega$ near Γ_2 or Γ_3 , with a strip width ϵ . I is the indicator function. $L(t)$ can be seen as a measure of the amount of time that the SRBM spends near the Neumann and Robin boundaries [24].

Therefore, the localized solution of the steady-state heat equation (6) can be converted into the probabilistic solution (7), and then be approximated by the Monte Carlo method with massive repeated SRBM path simulations to obtain the pointwise temperature solution.

D. Fixed and Floating Random Walk

In each Monte Carlo simulation, the SRBM path can be approximated by the random walk path. There are mainly two kinds of random walk methods.

- 1) *Fixed Random Walk*: The walk with fixed step length and preselected mesh nodes [21]. The particle jumps from one grid node to adjacent nodes. This walk process is time-consuming and cannot handle complex boundary conditions.
- 2) *Floating Random Walk*: The walk with neither the step length nor the mesh points preassigned. In particular, the

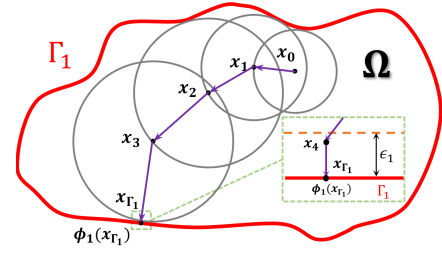


Fig. 3. Example of WOS with Dirichlet boundary.

WOS method [21], [32] employs maximum spherical transition regions inside the domain to greatly increase the speed of random walks.

Example of WOS: Considering a domain Ω governed by a Laplace equation with only Dirichlet boundary, as shown in Fig. 3, the probabilistic solution becomes [33]

$$T(\mathbf{x}) = E^{\mathbf{x}} \left(\phi_1(X_{t_{\Gamma_1}}) \right) \approx \frac{1}{N} \sum_{i=1}^N T(\mathbf{x}_{\Gamma_i}) \quad (10)$$

where $\mathbf{x}_{\Gamma_i}$ is the position when a path hits the Dirichlet boundary. i denotes the i th path of all N walking paths. $E^{\mathbf{x}}$ is the mathematical expectation.

A floating random walk procedure can be formulated to simulate each path. $T(\mathbf{x})$ can be obtained recursively through the following nested integral formula:

$$T(\mathbf{x}) = \oint_{S_1} P_1(\mathbf{x}, \mathbf{x}_1) T(\mathbf{x}_1) d\mathbf{x}_1 = \oint_{S_1} P_1(\mathbf{x}, \mathbf{x}_1) \oint_{S_2} P_2(\mathbf{x}_1, \mathbf{x}_2) \dots \oint_{S_{k+1}} P_{k+1}(\mathbf{x}_k, \mathbf{x}_{k+1}) T(\mathbf{x}_{k+1}) d\mathbf{x}_{k+1}, \dots, d\mathbf{x}_2 d\mathbf{x}_1 \quad (11)$$

where S_i , ($i = 1, \dots, k+1$) is the surface of the i th transition domain. $P_i(\mathbf{x}_{i-1}, \mathbf{x}_i)$, ($i = 1, \dots, k+1$) is the surface Green's functions relating the temperatures at $\mathbf{x}_{i-1}$ to $\mathbf{x}_i$.

Therefore, the WOS method employing spherical transition domains with variable radii can be constructed to calculate the temperature solution of each path. During the WOS process, given a starting point $\mathbf{x}_0$ in Ω , we can draw the largest sphere centered at $\mathbf{x}_0$ that does not intersect with the boundary $\partial\Omega$, and uniformly sample the second location $\mathbf{x}_1$ from the sphere surface. Next, take $\mathbf{x}_1$ as the new starting point, and perform the same spherical sampling. Repeat these operations until the path is absorbed by the Dirichlet boundary. The absorption occurs when the particle jumps to a location, such as $\mathbf{x}_4$ in Fig. 3, where the distance to Γ_1 is less than ϵ_1 . Then we record $\phi_1(\mathbf{x}_{\Gamma_1})$ for accumulation, where $\mathbf{x}_{\Gamma_1}$ is the nearest point to $\mathbf{x}_4$ on Dirichlet boundary. After simulating numerous paths, the numerical solution of $T(\mathbf{x})$ in (10) is acquired.

III. PATH INTEGRAL RANDOM WALK SOLVER FOR TRANSIENT THERMAL ANALYSIS

In this section, following the idea of steady-state analysis in Fig. 2, we present the probabilistic solution for the transient

thermal problem (1) based on the Feynman–Kac formula. The solution can then be simulated using the Monte Carlo method with random walks. Furthermore, we illustrate the novel techniques to perform random walks in both time and spatial dimensions, and handle mixed boundary conditions, especially Robin boundary conditions, and the heat-source regions.

A. Probabilistic Solution for Transient Heat Equation

For the transient heat equation (1), a more general form of the Feynman–Kac formula is leveraged, and the probabilistic solution is given as [29], [34]

$$T(\mathbf{x}, \tau) = E^{\mathbf{x}} \left\{ \hat{e}_c(t_{\Gamma_1}) \phi_1(X_{t_{\Gamma_1}}, t_{\Gamma_1}) \right\} \quad (12a)$$

$$+ E^{\mathbf{x}} \left\{ \int_0^{\tau} \hat{e}_c(t) \phi_{2,3}(X_t, t) dL(t) \right\} \quad (12b)$$

$$+ E^{\mathbf{x}} \left\{ \int_0^{\tau} \hat{e}_c(t) f(X_t, t) dt \right\} \quad (12c)$$

$$+ E^{\mathbf{x}} \left\{ \hat{e}_c(t_{\tau=0}) T_{\text{ini}}(X_{t_{\tau=0}}) \right\} \quad (12d)$$

where the temperature, boundary conditions, and the heat generation term are time-dependent as in (1)–(3). In addition to the contributions of Dirichlet boundary (12a), Neumann and Robin boundary (12b) and internal heat generation (12c), the initial condition contribution (12d) is also part of the final temperature solution. $t_{\tau=0}$ denotes the time when τ decreases to 0. Note that $\hat{e}_c(t)$ from Robin boundary conditions will influence all the path integrals as an attenuation coefficient.

Similarly, the probabilistic expression (12) can be discretized and approximated by the Monte Carlo method with massive repeated SRBM path simulations. The discrete form can be calculated as

$$\tilde{T}(\mathbf{x}, \tau) = \frac{1}{N} \sum_{i=1}^N \left\{ \hat{e}_c(t_{\Gamma_1}) \phi_1(X_{t_{\Gamma_1}}, t_{\Gamma_1}) \right\} \quad (13a)$$

$$+ \frac{1}{N} \sum_{i=1}^N \left\{ \sum_{s=0}^{NP_i} \hat{e}_c(t_j) \phi_{2,3}(X_{t_j}, t_j) dL(t_j) \right\} \quad (13b)$$

$$+ \frac{1}{N} \sum_{i=1}^N \left\{ \sum_{s=0}^{NP_i} \hat{e}_c(t_p) f(X_{t_p}, t_p) \Delta\tau \right\} \quad (13c)$$

$$+ \frac{1}{N} \sum_{i=1}^N \left\{ \hat{e}_c(t_{\tau=0}) T_{\text{ini}}(X_{t_{\tau=0}}) \right\} \quad (13d)$$

$$\hat{e}_c(t_j) = e^{\sum_{k=0}^j c(X_k) dL(t_k)} \quad (13e)$$

where N is the total number of random paths, s denotes each step of one path, and NP_i is the maximum number of steps of the i th path. In the Neumann and Robin boundary contribution (13b), j denotes the steps inside D_ϵ . In the heat-source region contribution (13c), p denotes the steps in the heat-source region, and $\Delta\tau$ is the fixed time interval consumed by each step inside the heat-source region. $\hat{e}_c(t_j)$ is updated only when the path hits the Robin boundary, gradually decaying from 1 to 0, and it affects all the temperature contributions (13a)–(13d).

Unlike the steady-state analysis, where WOS is only performed in 3-D space, we adopt a *spacetime* random walk with the time-dimension walk appended to calculate the temperature of point $\mathbf{x}$ at time τ .

B. Spacetime WOS in Heatless Region

Fig. 4 conceptually illustrates the *spacetime* WOS scheme utilized in the heatless region for transient thermal analysis [21]. In steady-state situations, the WOS is limited only in spatial dimensions, from $\mathbf{x}_i$ to $\mathbf{x}_{i+1}$ in one step. But in transient situations, the time-dimension walk needs to be considered. Therefore, the *spacetime* WOS scheme is constructed with the time-reverse walk appended and each step is from $(\mathbf{x}_i, \tau_i)$ to $(\mathbf{x}_{i+1}, \tau_{i+1})$, where the *reverse* means the walk starts at time $\tau = \tau_0$ and gradually decreases until $\tau = \tau_0 - \sum_i \eta_i = 0$, i.e., $\tau_3 = 0$ as shown in Fig. 4. During the walk process, time and space are not independent of each other due to a probabilistic constraint between the step radius r_i of the spatial walk and the interval $\eta_i = \tau_{i+1} - \tau_i$ of the time walk.

At the i th step in 3-D space, we draw a sphere with the maximum radius r_i fully contained in the domain, and the temperature of the central point $\mathbf{x}_i$ at time τ_i in terms of the surface temperature at time $\tau_i - \eta$ is given by (14), as shown at the bottom of the next page, [21], [35], where φ and θ are the azimuthal and polar angles, respectively. As described in Section II-D, the double integral with respect to F and G represents a uniform sampling of the next position $\mathbf{x}_{i+1}$ on the sphere surface with radius r_i .

$H(\alpha\eta/r_i^2)$ in (14b) indicates the intrinsic probabilistic constraint between the spatial radius r_i and the time interval η , where the right-hand side is an analytic expression including infinite series. The integral with respect to H denotes a uniform sampling over $H(\alpha\eta/r_i^2)$. As shown in Fig. 5, H can be regarded as the cumulative distribution function (cdf) of $\alpha\eta/r_i^2$, allowing inverse transform sampling on η_i with H^{-1} , the inverse function of H . Specifically, we enter the ordinate of Fig. 5 with a generated random number $w_i \in [0, 1]$, pick off the corresponding abscissa value, and calculate the time interval as [21]

$$\eta_i = H^{-1}(w_i) \frac{r_i^2}{\alpha}. \quad (15)$$

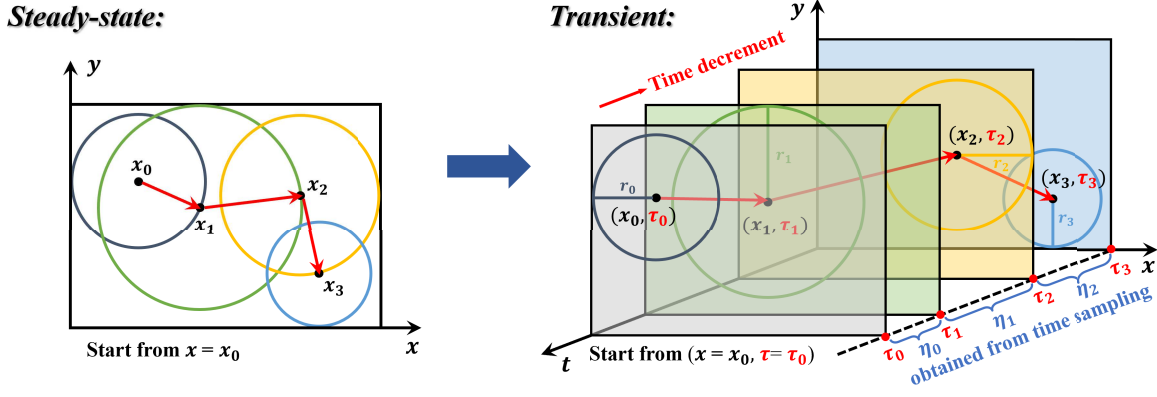
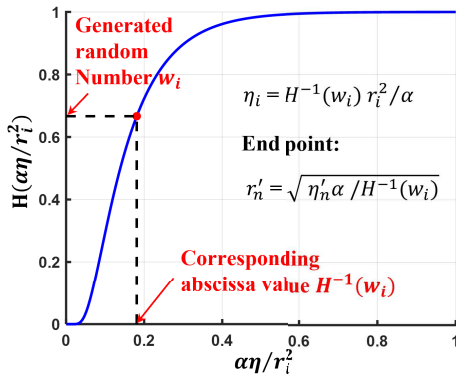
However, calculating the infinite series in H is time-consuming. In practice, we can precalculate and store the data points of $H(\alpha\eta/r_i^2)$ to improve computational efficiency.

When the time τ is close to 0, the operations of time termination adjustment are taken. If the current particle $(\mathbf{x}_n, \tau_n)$ finds $\tau_{n+1} = \tau_n - \eta_n < 0$, we manually select $\eta'_n = \tau_n$, and decrease the spatial radius as [21]

$$r'_n = \sqrt{\frac{\eta'_n \alpha}{H^{-1}(w_i)}} < r_n. \quad (16)$$

Then, the endpoint $\mathbf{x}'_{n+1}$ can be obtained through uniform sampling on a sphere with radius r'_n at $\tau_{n+1} = 0$, whose initial temperature $T_{\text{ini}}(\mathbf{x}'_{n+1})$ will be accumulated.

However, the primitive *spacetime* WOS in [21] cannot guarantee the accuracy of handling Robin boundary conditions, since it neglects the attenuated Feynman–Kac functional $\hat{e}_c(t)$ and adopts a difference-based treatment of probabilistic absorption or reflection at Robin boundary. To enhance accuracy, we take $\hat{e}_c(t)$ into consideration, so the initial temperature $T_{\text{ini}}(\mathbf{x}'_{n+1})$ of the endpoint should be weighted by $\hat{e}_c(t_{\tau=0})$ and accumulated in (13d). In addition, we employ a boundary

Fig. 4. Conceptual illustration of *spacetime* WOS scheme.Fig. 5. CDF $H(\alpha\eta/r_i^2)$ [21].

random walk mechanism near the Robin boundary to compute the boundary local time $L(t)$ and update $\hat{e}_c(t)$.

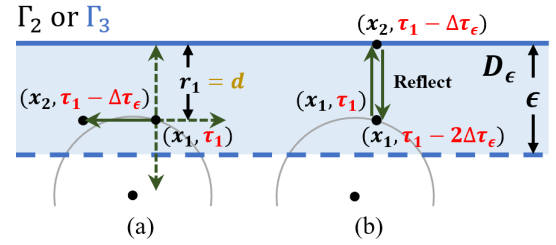
C. Treatment of Neumann and Robin Boundary

When the particle performs *spacetime* WOS in the heatless region, it may enter the D_ϵ strip region with width ϵ near the Neumann or Robin boundary, as illustrated by point x_1 at time τ_1 in Fig. 6. At this moment, it may affect the temperature according to (13b) and (13e), and therefore a special treatment near the boundary is necessary.

The boundary local time $L(t_j)$ in (13b) is also a stochastic process [29], [31]. In the fixed random walk, the increment

$$dL(t_j) \approx d \quad (17)$$

where d is the spatial step of the walking particle x [29], [31]. We apply it to the treatment within the boundary region D_ϵ , ensuring compatibility with the *spacetime* WOS strategy.

Fig. 6. Particle's behavior in D_ϵ region near Neumann or Robin boundary. (a) Particle transition within D_ϵ region. (b) Particle reflection upon hitting the boundary.

Specifically, to compute the boundary local time increment $dL(t_j)$, and update $\hat{e}_c(t_j)$ near the Robin boundary, we construct the following boundary *spacetime* random walk mechanism.

As shown in Fig. 6(a), similar to the fixed random walk [29], we restrict the particle's next jump to the six directions: $\pm x$, $\pm y$, and $\pm z$, and set the step length $d = r_1$, the distance from the particle to the boundary. Since the material is isotropic, the probability of the particle jumping in any of the six directions is 1/6. Each jump consumes a distance-dependent time interval of $\Delta\tau_\epsilon = d^2/6\alpha$ (to be derived in Section III-D). Therefore, the particle will arrive at point x_2 at time $\tau_1 - \Delta\tau_\epsilon$.

If the particle lands on the boundary [e.g., point x_2 in Fig. 6(b)], we consider the path hits the Neumann or Robin boundary. At this moment, it is as follows.

- 1) If it hits the Robin boundary, $\hat{e}_c(t_j)$ is updated according to (13e).
- 2) The contribution from the Neumann or Robin boundary is added to (13b) with $dL(t_j) \approx d$.
- 3) The particle will be reflected back to point x_1 and consume another time interval $\Delta\tau_\epsilon$, resulting in the particle still being at point x_1 at time $\tau_1 - 2\Delta\tau_\epsilon$.

$$T(x_i, \tau_i) = \int_{F=0}^1 \int_{G=0}^1 \int_{\eta=0}^{\tau_i} T(r_i, \theta, \varphi, \tau_i - \eta) dF(\varphi) dG(\theta) dH\left(\frac{\alpha\eta}{r_i^2}\right) \quad (14a)$$

$$F(\varphi) = \frac{\varphi}{2\pi}, \quad G(\theta) = \frac{1}{2}(1 - \cos\theta), \quad H\left(\frac{\alpha\eta}{r_i^2}\right) = 1 + 2 \sum_{k=1}^{\infty} (-1)^k \exp\left(-k^2 \pi^2 \frac{\alpha\eta}{r_i^2}\right) \quad (14b)$$

Experimental results show that after incorporating $\hat{e}_c(t)$ and boundary local time $L(t)$, and adopting the proposed boundary random walk mechanism, our method reduces the error of the primitive approach in [21] from 3.29 to 0.379 K in the chip cases without Dirichlet boundaries.

D. Time-Dependent Fixed Random Walk in Heat-Source Region

In order to handle single-layer or multilayer volumetric heat generation in transient problems, the fixed random walk with a fixed step time interval is developed in the heat-source region. We discretize the heat-source region into cuboid grids with even $\Delta x, \Delta y, \Delta z$. Utilizing the forward difference to approximate the time derivative and the second-order central difference to approximate the spatial derivative, the temperature at position $\mathbf{x} = (x, y, z)$ and time τ can be derived from (1) as [21]

$$\begin{aligned} T(\mathbf{x}, \tau) \approx & P_x^+ T(x + \Delta x, y, z, \tau - \Delta\tau) \\ & + P_x^- T(x - \Delta x, y, z, \tau - \Delta\tau) \\ & + P_y^+ T(x, y + \Delta y, z, \tau - \Delta\tau) \\ & + P_y^- T(x, y - \Delta y, z, \tau - \Delta\tau) \\ & + P_z^+ T(x, y, z + \Delta z, \tau - \Delta\tau) \\ & + P_z^- T(x, y, z - \Delta z, \tau - \Delta\tau) \\ & + P_0 T(x, y, z, \tau - \Delta\tau) + f(\mathbf{x}, \tau) \Delta\tau \end{aligned} \quad (18)$$

where $P_x^\pm = \alpha_x^\pm \Delta\tau / (\Delta x)^2$, $\alpha_x^\pm = k_x^\pm / c_p \rho$ denote the anisotropic thermal diffusivities in the $\pm x$ directions, where $k_x^\pm$ are the corresponding anisotropic thermal conductivities. $P_y^\pm, P_z^\pm$ are defined similarly. They are the probabilities of the particle jumping to the adjacent. $P_0 = 1 - (P_x^+ + P_x^- + P_y^+ + P_y^- + P_z^+ + P_z^-)$ is the probability to stay at the current node. Without loss of generality, we let $P_0 = 0$, and obtain the fixed time interval of each step as

$$\Delta\tau = \frac{(\Delta x \Delta y \Delta z)^2}{\alpha_z (\Delta x \Delta y)^2 + \alpha_x (\Delta y \Delta z)^2 + \alpha_y (\Delta z \Delta x)^2} \quad (19)$$

where $\alpha_x = \alpha_x^+ + \alpha_x^-$. α_y and α_z are defined similarly.

If the thermal conductivity k and thermal diffusivity α is isotropic, the fixed per-step time interval is

$$\Delta\tau = \frac{1}{2\alpha} \frac{(\Delta x \Delta y \Delta z)^2}{(\Delta x \Delta y)^2 + (\Delta y \Delta z)^2 + (\Delta z \Delta x)^2}. \quad (20)$$

Note that if $\Delta x = \Delta y = \Delta z = d$, then $\Delta\tau = d^2/6\alpha$. The $\Delta\tau_\epsilon$ consumed in each step of the boundary random walk in Section III-C corresponds to this special case.

Therefore, it is possible to construct a time-dependent fixed random walk for temperature calculation as follows. As shown in Fig. 7, assuming the particle is at location $\mathbf{x}_1$ and time τ_1 , it jumps to $\mathbf{x}_2$ with a possibility of P_x^- , and a reward of $\hat{e}_c(t_j)f(\mathbf{x}_1, \tau_1)\Delta\tau$ is accumulated in (13c). Similarly, the particle can then jump to $\mathbf{x}_3, \mathbf{x}_4, \dots$, until $\tau_1 - \sum \Delta\tau \leq 0$ or hits Dirichlet boundary.

Considering the interface of the heat-source region and heatless region, as illustrated in Fig. 8, we construct a virtual grid layer in the heatless area (white) close to the heat-source region (gray) [23]. When the particle enters the virtual layer at

Top view of heat-source region

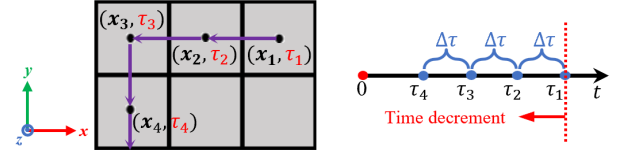


Fig. 7. Particle's behavior in the heat-source region.

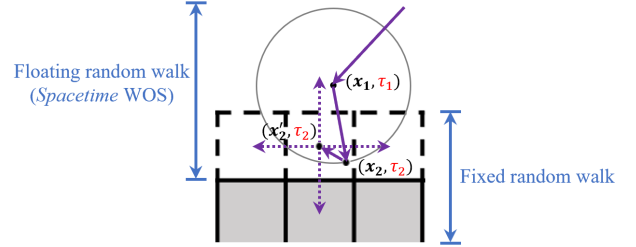


Fig. 8. Switch between fixed random walk and WOS [23].

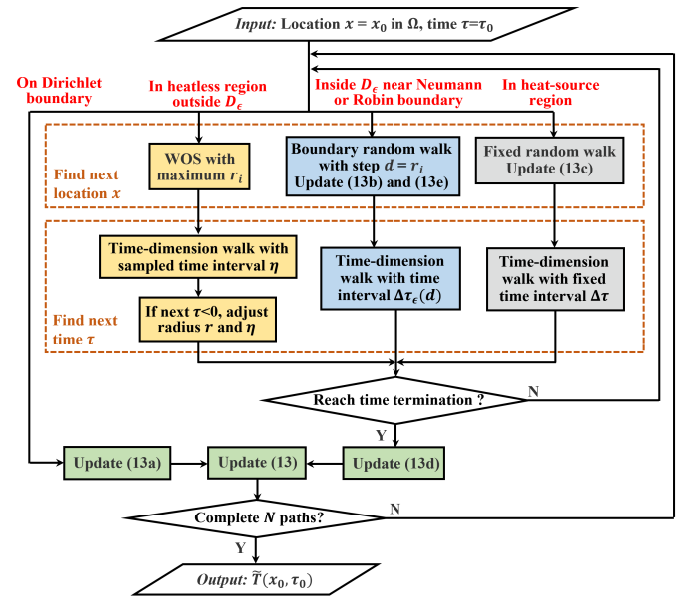


Fig. 9. Framework of proposed solver.

$\mathbf{x}_2$ at τ_2 , we pull it to the grid center $\mathbf{x}'_2$. Then, with assigning power $p(i) = 0$ to all nodes in the virtual layer, the particle executes a time-dependent fixed random walk.

E. Framework of Proposed Solver

The framework of our solver is summarized in Fig. 9. The solver takes the position $\mathbf{x}_0$ and the transient time τ_0 as input, execute N paths, and outputs the transient temperature $\tilde{T}(\mathbf{x}_0, \tau_0)$. Depending on the location of the particle, different strategies are utilized to find the next location $\mathbf{x}$, including the common WOS in heatless regions, the boundary random walk on D_ϵ near Neumann or Robin boundary, and the fixed random walk in heat-source regions. To find the next time τ , the time-dimension walk is executed with sampled time intervals in heatless regions, distance-dependent time intervals in D_ϵ regions, and a fixed time interval in heat-source regions.

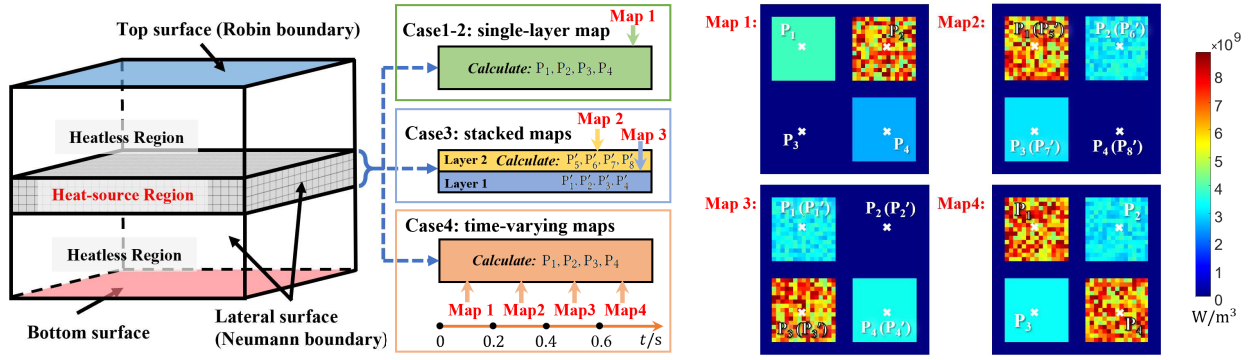


Fig. 10. Chip thermal model, case setting, and top view of power-maps.

Each path can be terminated by the time boundary τ_0 or by a Dirichlet boundary.

The accuracy and speed of stochastic methods are influenced by the number of paths N . As an unbiased Monte Carlo solver, the variance of the proposed solver is inversely proportional to path number N , according to the central limit theorem [26].

IV. EXPERIMENTAL RESULTS

This section evaluates the accuracy, efficiency, generalization, and parallelism of our transient path integral random walk (tPIRW) solver in several cases. All stochastic solvers are implemented in C++. All experiments are carried out on a Linux workstation with 80-core Intel Xeon CPUs of 2.30-GHz frequency and 376 GB of memory.

A. Accuracy and Efficiency Analysis of Chip Cases

To evaluate the accuracy improvement of tPIRW over the existing transient stochastic method, we reproduce the floating random walk method for transient problems from [21]. To adapt it to the heat-source region in chip cases, we integrate the fixed *spacetime* random walk from Section III-D, creating the hybrid method tHRW for subsequent comparison.

The primary difference between tHRW and tPIRW lies in their handling of $\hat{e}_c(t)$ in the path integral and Robin boundary conditions. Although both methods adopt *spacetime* WOS, tHRW neglects the attenuated $\hat{e}_c(t)$ and instead employs a probabilistic absorption-reflection approach at the Robin boundary. Note that we do not compare tPIRW with [27] as it addresses the coupled heat conduction and thermal radiation of structures under vacuum conditions without internal heat generation and convection, which differs from thermal issues of ICs.

The stochastic tPIRW solver is also compared with the deterministic methods, including ANSYS, the *de facto* commercial FEM simulator, and the CTM tool 3D-ICE, the state-of-the-art FDM-based thermal simulation tool for nano-scale CMOS technologies.

1) *Experimental Settings*: Without loss of generality, we model the geometry of a $2\text{ cm} \times 2\text{ cm} \times 1.1\text{-mm}$ chip as three stacked rectangular cuboids as shown in Fig. 10. The middle flat gray cuboid is the discretized heat-source region with a

thickness of $100\text{ }\mu\text{m}$, $k = 125\text{ W/(K}\cdot\text{m)}$, $c_p = 712\text{ J/(kg}\cdot\text{K)}$, and $\rho = 2330\text{ kg/m}^3$.

The top and bottom cuboid are heatless regions with a thickness of $500\text{ }\mu\text{m}$, $k = 395\text{ W/(K}\cdot\text{m)}$ and the same c_p, ρ . Lateral four surfaces are fixed as adiabatic Neumann boundary ($\partial T / \partial \vec{n} = 0$). The top surface is Robin boundary with $h = 8700\text{ W/(K}\cdot\text{m}^2)$ for cooling fans and heat sinks, and ambient temperature $T_{\text{amb}} = 318.15\text{ K}$ ($45\text{ }^\circ\text{C}$). The bottom surface is a Dirichlet boundary with $T = 343.15\text{ K}$ ($70\text{ }^\circ\text{C}$) or an adiabatic Neumann boundary. The initial temperature $T_{\text{ini}} = 318.15\text{ K}$ ($45\text{ }^\circ\text{C}$).

In stochastic methods, the number of paths $N = 5e^3$, and increased to $5e^4$ for the Dirichlet case with rapid temperature variation. The heat-source region is discretized into a $200 \times 200 \times 4$ rectangular grid for the fixed *spacetime* random walk. In tPIRW, the absorption strip width $\epsilon_1 = 1e^{-8}\text{ m}$, and the D_ϵ width ϵ is set equal to the height of a heat-source grid cell, i.e., $2.5e^{-5}\text{ m}$.

We randomly generate several 3-D volumetric power-maps (Map1–4) of four-core chips and impose them on the heat-source region as shown in Fig. 10. Four cases with different power-maps and boundary conditions on the bottom surface are evaluated as follows.

- 1) *Case 1*: Map1 and isothermal Dirichlet boundary on bottom.
- 2) *Case 2*: Map1 and adiabatic Neumann boundary on bottom.
- 3) *Case 3*: Map2 and Map3 to upper and lower heat-source layers, respectively, and adiabatic Neumann boundary on bottom.
- 4) *Case 4*: Map1–4 for 0.2 s in sequence to simulate workload switches and adiabatic Neumann boundary on bottom.

The bottom Dirichlet boundary in case 1 is an idealized approximation to showcase that the tPIRW solver accurately handles all three types of boundary conditions, and the adiabatic Neumann boundary on the bottom of cases 2–4 is more consistent with the realistic chip packaging. Cases 3 and 4 can verify the ability of the tPIRW solver to handle stacked 3-D ICs and chips with actual frequent switching workloads, respectively.

The solutions of the ANSYS simulator with very fine discretization (ANSYS_G) are taken as the golden reference.

TABLE I
ACCURACY AND EFFICIENCY OF ANSYS, 3D-ICE, tHRW, AND tPIRW

Case	Solver	Elems. Size (m)	# Elems.	Time Step Size (s)	# Time Steps	Avg. Abs. Err. (K)	Max. Abs. Err. (K)	Time*(s)	Speedup	Mem.*(GB)	Mem. Red.
1	ANSYS_G	8e-5	6.04e6	5e-5	400	-	-	5468	-	33.2	-
	ANSYS_C	1e-4	3.80e6	5e-4	40	0.287	1.15	470	0.636	20.2	0.193
	3D-ICE	1e-4	4.40e5	5e-4	40	0.275	1.36	299	1.00	3.9	1.00
	tHRW	-	-	-	-	0.063	0.198	0.186	1608	6.6e-3	591
	tPIRW	-	-	-	-	0.049	0.207	0.123	2431	6.6e-3	591
2	ANSYS_G	8e-5	6.04e6	1e-3	1e3	-	-	13060	-	33.8	-
	ANSYS_C	1e-4	3.80e6	0.01	100	0.116	0.433	958	0.328	20.5	0.190
	3D-ICE	1e-4	4.40e5	0.01	100	0.057	0.292	314	1.00	3.9	1.00
	tHRW	-	-	-	-	1.22	3.29	1.90	165	4.8e-3	813
	tPIRW	-	-	-	-	0.099	0.379	1.79	175	4.8e-3	813
3	ANSYS_G	8e-5	7.64e6	1e-3	1e3	-	-	29466	-	44.1	-
	ANSYS_C	1e-4	4.99e6	0.01	100	0.096	0.277	1152	0.330	28.5	0.151
	3D-ICE	1e-4	4.80e5	0.01	100	0.097	0.271	380	1.00	4.3	1.00
	tHRW	-	-	-	-	1.22	3.02	1.95	195	5.2e-3	827
	tPIRW	-	-	-	-	0.083	0.337	1.80	211	5.4e-3	796

* According to Sec. IV-A3, the time and speedup are for transient time 0.01s (case 1) and 0.5s (cases 2-3), as are the memory and memory reduction.

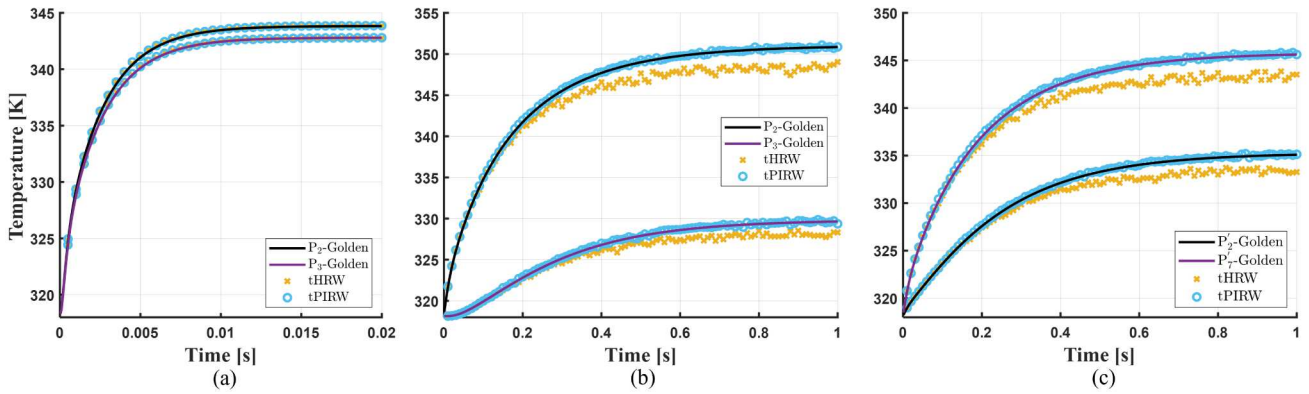


Fig. 11. Accuracy of partial interested points between stochastic methods. (a) Solutions of P_2 and P_3 in case 1. (b) Solutions of P_2 and P_3 in case 2. (c) Solutions of P'_2 and P'_7 in case 3.

To fairly compare the speed, we construct coarse FEMs, called ANSYS_C. The element sizes of deterministic methods ANSYS_C and 3D-ICE are the same as the grid size of the heat-source region in tPIRW. Since the original 3D-ICE does not support Dirichlet boundary directly, we approximate it by setting a very large heat transfer coefficient $h = 1e^{12}$ W/(K·m²) on Robin boundary.

Frankly, for speed comparison, simply comparing runtime is unfair to deterministic methods, as they solve the entire temperature field, while stochastic algorithms compute only point-wise temperatures. However, since stochastic methods are typically used to evaluate points of interest in iterative optimization scenarios, extremely fast evaluation is crucial for shortening optimization cycles. Thus, we adopt a simplified comparison based on their runtimes. For multipoint cases, the inherent parallelism of Monte Carlo methods allows latency-free acceleration on multicore CPUs or GPUs [36].

For stochastic methods, we assume the interested points (P_1 – P_4 in cases 1, 2, and 4 and P'_1 – P'_8 in case 3) are the centers of cores as shown in Fig. 10. In Table I, the “Time” for tPIRW and tHRW refers to the average time to compute the transient temperature at a single point, while for ANSYS and 3D-ICE,

it represents the time for a full-chip simulation, as they cannot obtain point-wise results without solving the entire domain.

2) *Accuracy Comparison:* Although tHRW can provide decent accuracy in case 1 as shown in Fig. 11(a), when Dirichlet boundary is excluded in cases 2 and 3, their solutions deviate significantly from the golden ones, as shown in Fig. 11(b) and (c). Table I shows that the maximum absolute error of tHRW is 3.29 K, which is undesirable in engineering applications. More seriously, the case-by-case accuracy instability observed will undermine engineers’ confidence in the method. In contrast, tPIRW can maintain high accuracy with absolute errors no more than 0.4 K in all cases, regardless of different boundary conditions and power maps.

The improved accuracy of tPIRW over tHRW is attributed to the introduction of the Feynman–Kac functional $\hat{e}_c(t)$ and the boundary local time $dL(t)$ in the path integral, along with the corresponding boundary random walk mechanism proposed in Section III-C, rather than the difference-based scheme of probabilistic absorption or reflection at the Robin boundary.

In deterministic methods, ANSYS_C and 3D-ICE offer acceptable accuracy with average absolute error within 0.3 K and maximum absolute error within 1.4 K. Compared with

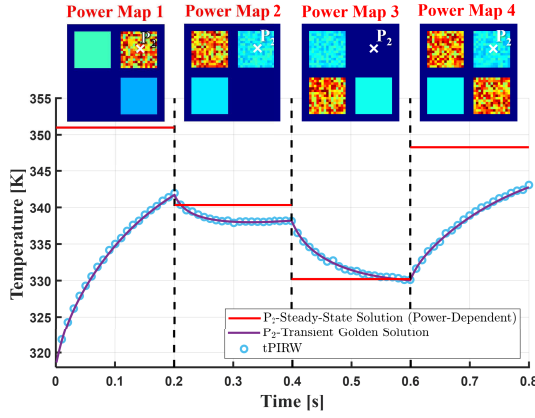


Fig. 12. Accuracy under frequent switching workloads (Solutions of P_2 in case 4). “ P_2 -Steady-State Solution (Power-Dependent)” denotes the imaginary steady-state temperature of P_2 under the current applied power map (e.g., Map1 for 0–0.2 s, Map2 for 0.2–0.4 s).

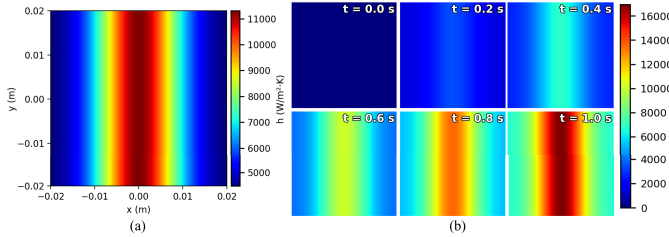


Fig. 13. Distribution of the variable convective heat transfer coefficient $h(x,t)$ at Robin boundary. (a) Varying in space: $h(x,t) = 8700(0.5 + 0.8 \exp(-10000x^2))$. (b) Varying in both space and time: $h(x,t) = 13050(0.5 + 0.8 \exp(-10000x^2))$.

these deterministic methods, tPIRW achieves better accuracy in case 1 with errors within 0.3 K, and comparable accuracy in cases 2 and 3 with errors less than 0.4 K.

Fig. 12 shows the temperature trace of a point under rapid workload switches. Since the switching interval is 0.2 s, much smaller than the thermal relaxation time, the gap between transient solutions and steady-state solutions (red lines) may be very large, such as more than 10 K in the first 0.2 s in Fig. 12. Therefore, transient thermal analysis for modern chips is necessary. As a precise transient solver, tPIRW can accurately obtain the transient temperature with an average error of 0.13 K and a maximum error of 0.54 K.

To emulate a more realistic cooling scenario, we vary the heat transfer coefficient $h(x,t)$ at the top Robin boundary in both space and time (in case 2), and report the accuracy of tPIRW relative to ANSYS_G. We construct two distributions of $h(x,t)$ in Fig. 13. In Fig. 13(a), $h(x,t)$ follows a Gaussian profile along the x -direction, approximating a forced-air heat sink with a peak centerline velocity that decays laterally toward the sides. In Fig. 13(b), $h(x,t)$ retains the above spatial profile and increases over time to simulate the heat sink’s start-up process.

Under the $h(x,t)$ distribution in Fig. 13(a), tPIRW attains a maximum absolute error of 0.36 K; under Fig. 13(b), the maximum absolute error is 0.59 K. The accuracy at partial points of interest is shown in Fig. 14. These results demonstrate that tPIRW maintains high accuracy with a varying $h(x,t)$.

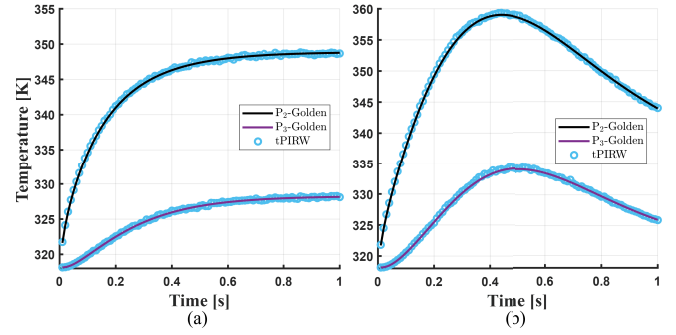


Fig. 14. Accuracy of interested points P_2 and P_3 in case 2 for the variable heat transfer coefficient $h(x,t)$. (a) Varying in space. (b) Varying in both space and time.

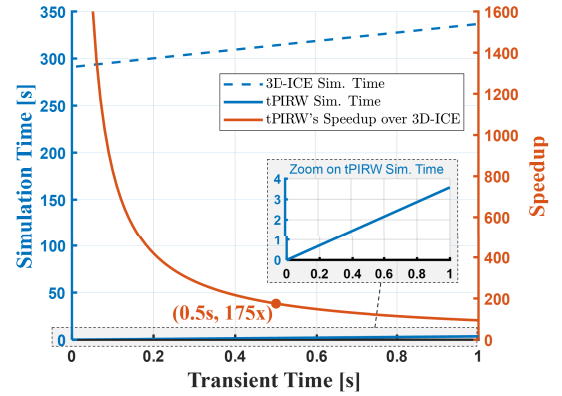


Fig. 15. tPIRW’s speedup over 3D-ICE in case 2 for single-point thermal simulation.

3) *Speed and Memory Comparison*: Deterministic methods such as FEMs and FDMs convert the heat equations into sets of ordinary differential equations and solve them using numerical integration with standard techniques such as Euler’s method or the Runge–Kutta method. Therefore, it is necessary to solve the corresponding linear equations at each time step. In engineering practice, methods like 3D-ICE and ANSYS save the matrix decompositions, LU, for example, to facilitate reuse in subsequent time steps, thereby improving the speed of transient solutions.

As shown in Fig. 15, the simulation time of 3D-ICE is linearly related to the required transient time with an intercept because it needs time to construct and process corresponding thermal resistance, thermal conductivity, and power matrices. The simulation time of tPIRW is proportional to the required transient time. Therefore, when the required transient time is small, the speedup of tPIRW over 3D-ICE can reach up to $1600\times$. However, for a fair comparison, the simulation times and speedups of different methods in Table I are given at transient time 0.01 s (case 1) or 0.5 s (cases 2 and 3), which are close to the thermal relaxation times.

As shown in Table I, 3D-ICE achieves $1.6\text{--}3.1\times$ speedup and $5.2\text{--}6.6\times$ memory reduction relative to ANSYS_C, as the latter uses tetrahedral meshes, which have more elements compared to the cubic meshes of 3D-ICE. In the idealized case 1, for single-point temperature calculation, our tPIRW achieves $2431\times$ speedup and $591\times$ memory reduction over

TABLE II
ACCURACY AND EFFICIENCY OF ANSYS, 3D-ICE, AND tPIRW FOR A 2.5-D-IC SYSTEM

Case	Solver	Elems. Size (m)	# Elems.	Time Step (s)	# Time Steps	Avg. Abs. Err. (K)	Max. Abs. Err. (K)	Time*(s)	Speedup	Mem. *(GB)	Mem. Red.
Compact	ANSYS_G	2e-4	2.00e6	5e-4	800	-	-	4375	-	12.2	-
	ANSYS_C	2.5e-4	1.15e6	5e-3	80	0.117	0.441	366	0.058	7.2	0.106
	3D-ICE	2.5e-4	1.79e5	5e-3	80	0.133	0.346	21.4	1.00	0.76	1.00
	tPIRW	-	-	-	-	0.267	0.605	0.73	29.3	4.3e-3	177
Loose	ANSYS_G	2e-4	1.99e6	5e-4	800	-	-	4786	-	12.2	-
	ANSYS_C	2.5e-4	1.14e6	5e-3	80	0.120	0.428	369	0.057	7.2	0.106
	3D-ICE	2.5e-4	1.79e5	5e-3	80	0.089	0.331	21.2	1.00	0.76	1.00
	tPIRW	-	-	-	-	0.158	0.563	0.79	26.8	4.3e-3	177

* According to Sec. IV-A3, the time and speedup are for transient time 0.2s, as are the memory and memory reduction.

3D-ICE with the help of Dirichlet boundary absorption. In realistic cases 2 and 3 without the Dirichlet boundary, tPIRW still maintains a speedup of over $175\times$ and over $796\times$ memory reduction compared to 3D-ICE. Between stochastic methods, as shown in Table I, tPIRW is slightly faster than tHRW and consumes a similar amount of memory.

The remarkable speedup and memory reduction over deterministic methods come from the fact that, when only a few points of interest need to be solved, the stochastic solver tPIRW avoids the full spatial discretization, time discretization, linear system formulation, and large matrix inversion required by traditional deterministic methods.

Note that steady state is the $t \rightarrow \infty$ special case of the transient problem, and hence a steady-state result can be obtained by running a transient solver for a sufficiently long time. This is precisely how the prior steady-state solver [23], [26] operates: it continues the random walk until the temperature solution converges. Our transient solver tPIRW terminates trajectories early once the required transient time τ_0 is reached. Consequently, the tPIRW transient solver is usually faster than the previous steady-state solver.

B. Complex Structure Validation via 2.5-D-IC Systems

The tPIRW, as a general field solver for the heat equation (diffusion equation), can be applied to transient heat analysis for complex geometric structures and boundary conditions without special treatment. In this subsection, we conduct transient thermal analysis on a 2.5-D-IC system with chiplets and present the comparative results with other deterministic methods.

The modeling of the 2.5-D-IC system is illustrated in Fig. 16. The passive interposer in Fig. 16(a) is modeled as the bottom layer cuboid in Fig. 16(b) with an adiabatic (Neumann) boundary at the interface to the C4 pillar. The heat sink and heat spreader are modeled as the top layer cube with a convective (Robin) boundary at the interface to the environment. The chiplets correspond to the heat source regions, and the epoxy is modeled as adiabatic regions due to low thermal conductivity $[0.1\text{--}0.4 \text{ W/(K}\cdot\text{m)}]$. Different from the settings in Section IV-A, the thermal model dimensions are $4 \text{ cm} \times 4 \text{ cm} \times 0.7 \text{ mm}$, with layer thicknesses and thermal conductivities of 100, 100, and $500 \mu\text{m}$, and 125, 125, and $395 \text{ W/(K}\cdot\text{m)}$ from bottom to top, and the convective heat transfer coefficient h is $15000 \text{ W/(K}\cdot\text{m}^2)$. In tPIRW, the heat

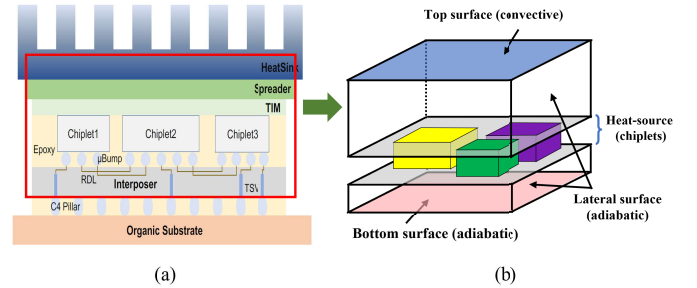


Fig. 16. Modeling process of the 2.5-D-IC system. (a) Structure of 2.5-D-IC systems [37]. (b) Thermal model.

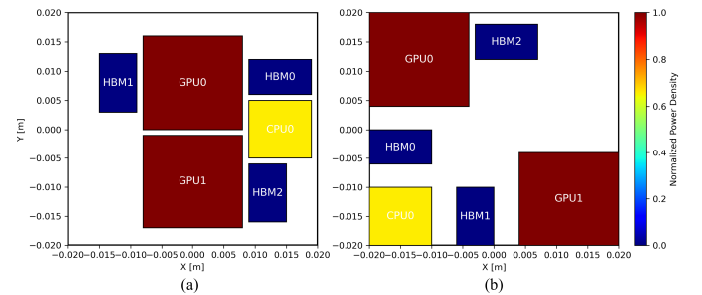


Fig. 17. Power-maps of the two chiplet placement systems (maximum power density = 5.86e9 W/m^3 , minimum power density = 3.33e9 W/m^3). (a) Compact case. (b) Loose case.

source region is discretized into a $160 \times 160 \times 4$ grid. Other settings match those in Section IV-A.

We adopt two typical chiplet placements from [37] and [38] as test cases: a compact and a loose arrangement of a 2.5-D-IC system consisting of two GPUs, one CPU, and three HBMs. The resulting power maps are shown in Fig. 17, with the center of each chiplet taken as the points of interest for measurement. Similarly, the “Time” for tPIRW in Table II refers to the average simulation time per point. We find that the points’ temperature reaches a steady state around 0.2 s. Therefore, similar to Section IV-A, we perform a 0.4-s simulation and present the simulation times for each method at 0.2 s.

Note that the newest available 3D-ICE v3.1 does not currently provide an interface for assigning different thermal conductivities k to cells within the same layer. Therefore, we modify its source code to add this functionality, enabling 3D-ICE to support thermal simulation of 2.5-D IC systems with chiplets.

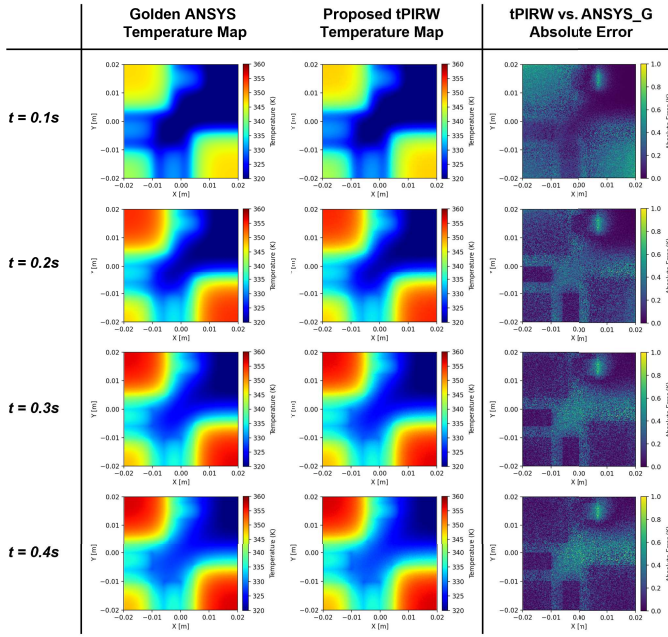


Fig. 18. Accuracy of tPIRW on the global temperature map at 0.1, 0.2, 0.3, and 0.4 s in the loose case in Fig. 17(b).

As shown in Table II, the maximum error of 3D-ICE is 0.346 K, and the maximum error of ANSYS_C is 0.441 K. In this case, tPIRW maintains comparable accuracy with a maximum error within 0.61 K, and achieves a $26.8\times$ single-point speedup compared to the faster deterministic method, 3D-ICE. The experimental results validate the general geometry applicability of tPIRW.

The speedup of tPIRW over 3D-ICE is lower in this test case compared to the chip cases. This is because the discretization in this example is relatively coarse, with fewer elements, making matrix LU decomposition and time stepping in deterministic methods relatively fast. Therefore, it is reasonable to infer that tPIRW offers greater speed advantages over deterministic methods in larger scale chip test cases.

Fig. 18 shows the global temperature map of Fig. 17(b). Similar results can be obtained for Fig. 17(a). We calculate the temperatures of points on a 160×160 grid. The average absolute error is 0.2 K, and the maximum absolute error is 1 K, which verifies the correctness of the stochastic method tPIRW as a field solver for the full field.

C. Parallelism

In the experiments of the first two subsections, the ANSYS simulator is run with 16-core parallelism, as we find that further increasing the number of cores yields little or even negative speedup, and significantly increases memory consumption. The open-source 3D-ICE tool is single-threaded. Although increasing its thread count can further improve performance, experiments in [8] show that the speedup plateaus at around four threads.

In contrast, stochastic methods like tPIRW have high parallel efficiency, enabling significant single-point speedup over deterministic methods on multicore systems. The excellent

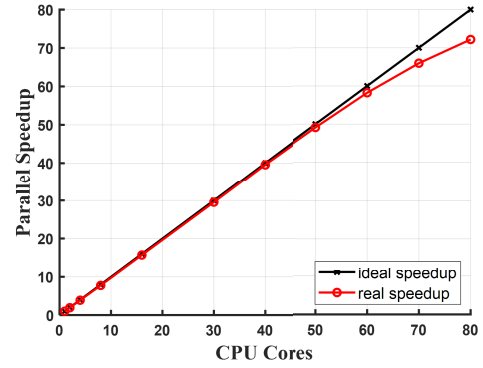


Fig. 19. Parallelism of tPIRW solver.

parallel efficiency of tPIRW is attributed to the intrinsic parallelism of Monte Carlo methods, namely the high independence of paths and the similarity in particle behavior within each path. We conduct this experiment under the conditions of case 2 in Section IV-A, using parallel CPU cores ranging from 1 to 80. A customized thread-safe program is utilized to facilitate multithreaded acceleration.

Fig. 19 shows that as the number of CPU cores increases, the speedup of tPIRW maintains an almost ideal linear growth with more than 90% parallel efficiency, demonstrating significant advantages of tPIRW in large-scale parallel computing.

V. CONCLUSION

In this article, an accurate and efficient stochastic solver is developed for IC transient thermal analysis based on the Feynman–Kac formula, which handles all three types of boundaries with high precision, and accommodates single-layer and multilayer volumetric power-maps. The proposed solver maintains high accuracy with average absolute error within 0.27 K and maximum absolute error within 1 K, achieves $26.8\text{--}211\times$ speedup and $177\text{--}813\times$ memory reduction at a single point on realistic cases compared with the state-of-the-art deterministic solver 3D-ICE, and exhibits high parallelism capabilities. As a field solver, its general applicability is also validated through the example of a 2.5-D-IC system.

Although tPIRW, as a localized solver, shows strong potential for fast local thermal analysis in the early stages of large-scale chip design, its applicability can be further extended to fusion with deterministic and NN-based methods to form a global solver. Meanwhile, future research work may include temperature-dependent thermal conductivity, imperfect interfacial contact, and microscopic thermal transport.

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